

A Risk Analysis of the Effects of Coral Reef Degradation in Tropical Climates

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CAAARal Reefs

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Executive Summary

Coral reef degradation is currently at an alarmingly high rate, with nearly half of global hard coral cover lost and degradation approaching 70% for certain reefs. Over half a billion people directly rely on reefs for their livelihoods and food, and the effects extend well beyond subsistence. Coral reefs absorb 97% of wave energy from storms, support fisheries and tourism generating approximately \$3.4 billion in global economic value annually, and sustain roughly twenty-five to thirty-two percent of all marine species. The primary drivers of reef degradation are rising ocean temperatures, ocean acidification, and greater cyclone frequency, all of which are projected to intensify in the future. Critically, reef degradation also imposes irreversible pharmaceutical losses: four reef-derived compounds have reached clinical approval to date, and species driven to extinction before discovery are permanently removed from the pharmaceutical pipeline regardless of subsequent restoration. Low-income and indigenous communities are disproportionately affected by degradation across all channels, and the downstream impacts on fishing, tourism, coastal protection, and biodiversity threaten both local economies and global markets. This paper presents a quantitative risk analysis of coral reef degradation across seven global reef regions and proposes a set of government programs to mitigate these risks.

Data were drawn from over 1.6 million observations across four platforms — NOAA CORIS, IBTrACS, The World Bank, and FRED — spanning benthic cover, fish population surveys, coral bleaching, water temperature, and carbonate chemistry. Four models were integrated into a unified analytical pipeline. A stochastic Gordon-Schaefer Model projected fish biomass trajectories and collapse probabilities, finding that catches are decreasing across all regions, with total global fishing losses of \$4.6 billion by 2100 and collapse probabilities exceeding 97% at every location. A SARMA(1,1) $\times$ (1,1)₄ time-series model and SARIMAX extension projected a continued hard coral cover decline of -0.6397% per year, with episodic near-zero cover events emerging at eight-year intervals by the 2030s as rising algae cover, from 42.7% in 2013 to 65.2% in 2023, continues to displace live coral. An economic loss model translated these forecasts into annual losses across tourism and coastal wave damage channels, growing from \$892 million in 2025 to \$4.59 billion per year by 2100, with a 77-year net present value of \$93.07 billion under the status quo. Finally, a pharmaceutical option value model applied a bioeconomic prospecting framework across all seven regions, projecting \$172.6 billion in cumulative pharmaceutical NPV losses by 2100, with permanent species losses concentrated at episodic bleaching trough years in 2028, 2036, 2044, and beyond — losses that are irreversible regardless of subsequent reef restoration.

A sensitivity analysis across three intervention scenarios shows that holding current cover constant saves \$35.96 billion in NPV losses across fishery, tourism, and wave damage channels, while active restoration saves \$88.77 billion, confirming that reef restoration is a more critical next step than simply curbing current degradation. Pharmaceutical losses at bleaching trough years are not recoverable under either scenario, establishing an additional \$172.6 billion in value at risk that restoration alone cannot protect. Combined across all channels, total projected losses through 2100 reach approximately \$270 billion under the status quo.

Based on these findings, this paper recommends a six-program government policy package totaling \$1,953.5 million in annual government outlay (\$1,969 million total including a 50% public-private cost-share on Program 6), spanning all three categories of risk mitigation: modifying outcomes through a ten-year coral restoration program, which targets 96,171 hectares at \$200,000 per hectare as well as expanded Marine Protected Area enforcement; changing behavior through fishermen income subsidies and equipment rebates to support transitions away from reef-dependent fishing; and providing insurance through a parametric national reef reserve funded by fishing license and tourism levies to enable rapid response after shock events. A sixth program — a public-private pharmaceutical biodiversity prospecting fund — cryopreserves and screens reef-organism tissue samples before bleaching troughs foreclose discovery permanently, converting irreversible pharmaceutical option value losses into a durable banked asset. These initiatives are intended to guide reef restoration and biodiversity protection efforts before critical thresholds are reached by the mid-2050s.

Ultimately, this paper provides a framework for quantifying the impacts of reef degradation across fishery, tourism, wave damage, and pharmaceutical channels, developing integrated strategies to aid in restoration efforts, and recommending governmental policies to protect financial institutions and the millions relying on reefs for their livelihoods globally.

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1 Introduction & Background

Coral reefs, distributed across tropical and subtropical ocean floors worldwide, are among the most biologically and economically valuable ecosystems on Earth. Widely marvelled at for their aesthetic value, coral reefs play a pivotal role in facilitating interactions between species across ecosystems. In addition, coral reefs are directly responsible for supporting approximately twenty-five to thirty-two percent of all marine species despite covering less than one percent of the ocean floor [28, 31].

In addition to their ecological importance, coral reefs provide substantial economic value through fisheries, tourism, and natural coastal protection. Reefs absorb up to 97% of wave energy from storms, reducing erosion in coastal communities and providing flood risk protection totaling approximately \$1.8 billion. Globally, more than 1 billion people rely directly or indirectly on coral reef ecosystems for food security, employment, and income. Fisheries and tourism generate approximately \$200 million annually in reef-dependent regions, while the total global economic value of coral reef-related services is estimated at \$3.4 billion [15]. Coral reefs therefore represent one of the most integral environmental assets supporting global economies.

1.1 Causes and Risks

Despite their immense per-area benefit to marine life and humans alike, coral reefs face accelerating existential risks driven primarily by rising ocean temperatures, ocean acidification, and increased storm intensity. Elevated sea surface temperatures and intensified cyclones trigger mass coral bleaching events, while ocean acidification weakens coral skeletons, reducing reef resilience and recovery potential [9, 10, 27]. When simplified, the primary drivers of coral reef degradation are increasing global ocean temperatures, rising acidity levels, and greater cyclone frequency.

As these stressors intensify, coral reef loss poses systemic risks that extend beyond marine ecosystems, threatening food security, livelihoods, and economic stability. The resulting risks include sustained declines in reef-dependent fish populations, increased coastal exposure to storm damage, and long-term economic instability in communities and industries that rely on healthy coral reef systems. Furthermore, reef degradation imposes irreversible losses on the pharmaceutical sector. Coral reef organisms have yielded four clinically approved compounds to date — AZT (HIV/AIDS), Cytarabine (leukemia), Ziconotide (chronic pain), and Trabectedin (soft-tissue sarcoma) — and systematic screening programs suggest that reefs harbor bioactive chemistry at rates substantially above the marine average, owing to the intense competitive chemical ecology of high-density reef environments [17, 23]. Unlike tourism or fishery revenue, pharmaceutical option value lost to species extinction before discovery is permanently foreclosed: no subsequent restoration can recover a compound from a species that no longer exists [3]. This makes pharmaceutical biodiversity loss categorically distinct from the other risk channels analyzed in this paper and motivates a dedicated model and policy response in Sections 3.5 and 5.2.

These impacts are most concentrated in tropical and coastal regions, where reef degradation disproportionately affects fishermen, Indigenous communities, and low-income populations with limited economic alternatives [25]. Many Small Island Developing States (SIDS), whose economies rely heavily on reef-based tourism due to the aesthetic and ecological value of their coastal environments, are particularly vulnerable. Overall, the cascading effects of coral reef degradation can be categorized into two environmental levels that translate into broader socioeconomic consequences.

1.2 Effect Categorization

Table 1: Cascading Effects of Coral Reef Degradation with Socioeconomic Impacts

#	Level I Effect	Level II Effect	Socioeconomic Impacts
1	Coral reef area ↓	Fish populations ↓	Food supply & revenue ↓
2	Coral bleaching ↑	Reef aesthetic value ↓	Tourism revenue ↓

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Table 1 – *continued from previous page*

#	Level I Effect	Level II Effect	Socioeconomic Impacts
3	Hardbottom coral cover ↓	Wave energy absorption ↓	Tropical storm damage ↑

1.3 Existing Mitigation Strategies

Current coral reef mitigation strategies include establishing marine protected areas, expanding coral restoration and nursery programs, developing heat-tolerant coral strains, installing artificial reef structures, and implementing policies to reduce pollution while promoting reef-safe tourism practices.

While these approaches have demonstrated promising localized success, significant gaps remain in long-term, large-scale empirical evidence regarding their economic efficiency, scalability, and sustained ecological impact across varying climate conditions and geographic regions. Addressing these gaps requires integrated modeling approaches capable of linking environmental degradation to measurable socioeconomic risk.

1.4 Problem Statement

There is an imperative need for a foundational understanding of the risks involved with coral reef degradation in order to guide environmental policy. The primary risk factors driving this degradation include rising ocean temperatures, ocean acidification, and increased cyclone frequency, which together threaten the food security, livelihoods, and economic stability of reef-dependent communities. In this paper, we propose various regression-based models for analyzing the social and economic risks of reef degradation. Our project aims to understand changes in reef size, fish populations, ocean temperature, and acidity to assess the downstream social and economic impacts of reef loss. Our analysis is designed to support decision-making for key stakeholders, including fishermen, Indigenous and coastal communities, tourism-dependent industries, insurers, and regional governments. By identifying how environmental variables drive reef degradation and translating these effects into measurable economic and social risks, this paper aims to guide reef risk mitigation. Specifically, our findings are intended to aid policy development, insurance design, and conservation investment strategies that reduce long-term losses, enhance ecosystem resilience, and protect vulnerable populations who depend most heavily on healthy coral reef systems. To address these risks, this paper explores three categories of mitigation: direct outcome modification through reef restoration and a pharmaceutical biodiversity prospecting fund that banks species option value before bleaching troughs foreclose discovery permanently; behavioral change through subsidies and conservation policy; and insurance mechanisms to protect reef-dependent communities from acute losses. Six government programs spanning all three categories are proposed in Section 5, including a novel public-private cryopreservation program that converts at-risk pharmaceutical option value into a durable asset independent of reef cover trajectory.

2 Data Methodology

The data for this project spans four independent data platforms and incorporates a variety of environmental and socioeconomic metrics relevant to coral reef health and economic impacts. Data were sourced primarily from the National Oceanic and Atmospheric Administration’s Coral Reef Information System (CORIS) and the International Best Track Archive for Climate Stewardship (IBTrACS) [12, 16]. Economic data used for model calibration and validation were obtained from the World Bank and the Federal Reserve Economic Data (FRED) [4, 26]. Together, these four platforms provide over 1.6 million individual observations across 7 reef regions, forming a comprehensive basis for assessing reef degradation and related economic consequences.

2.1 Data Management and Cleaning

All datasets were managed in a GitHub repository, enabling version control and centralized storage. Data files were stored in .csv format and sorted by region (e.g., Puerto Rico, U.S. Virgin Islands, CNMI). Each

sub-region dataset contained multiple metrics, including benthic cover, fish population surveys, erosion rates, calcification rates, and carbonate chemistry measurements. Dataset sizes varied: reef and bleaching surveys contained at least 30 entries per file, while fish surveys included up to 50,000 entries. This provides a clear numerical summary of the data volume per category, supporting transparency in the modeling process.

Data cleaning was conducted in Jupyter Notebook using Python. Cleaning steps included the removal of columns with incomplete or non-informative data, the standardization of reef location identifiers for consistency across datasets, and verification of entries against source documentation to ensure accuracy. Survey years with no benthic observations for a given transect were excluded from the panel regression rather than imputed, to avoid introducing artificial cover values into the trend estimation. Entries with physiologically implausible values (e.g., combined benthic cover fractions exceeding 100%) were flagged and removed, accounting for less than 0.3% of total observations. The carbonate chemistry and SST records from CORIS, which are measured at higher temporal frequency than annual benthic surveys, were aggregated to annual means by reef region prior to joining with the survey-level data.

These steps ensure the reliability and credibility of the data while maintaining replicability. After cleaning, datasets were assessed for suitability for modeling and categorized according to the Actuarial Process Guide, as summarized in Table 2.

Five core metrics characterize reef degradation in this analysis: **erosion** (rate of reef structural loss, increasing coastal flood risk), **carbonate chemistry** (ocean pH conditions governing coral growth rates), **benthic cover** (percentage of seafloor covered by living coral, the primary indicator of reef health), **fish population surveys** (abundance and diversity of reef-dependent species), and **coral reef bleaching** (loss of symbiotic algae under thermal stress, reducing coral survival and reef functionality).

2.2 Data Assumptions and Justifications

The datasets were assessed for reliability, completeness, and applicability to the modeling objectives. Of the four source platforms, CORIS and IBTrACS provide direct observational measurements and are considered the most reliable inputs; World Bank and FRED data are secondary economic aggregates subject to revision and varying methodological consistency across countries and time periods [4, 12, 16, 26].

Data assumptions and justifications are as follows. First, time-series environmental and economic data are assumed to be continuous and to follow smooth trends; while short-term fluctuations may exist, the regression-based modeling framework captures overall trends and minimizes noise effects. Second, data accuracy is assumed to reflect current conditions, as all data were collected from reputable open-source sources and verified internally by CORIS and NOAA prior to use. Third, observed outliers are treated as real-world deviations rather than reporting errors, and over long time horizons the data are assumed to approximate expected statistical distributions. Fourth, Florida Keys economic parameters are taken as representative of U.S. reef regions more broadly; although the economic loss model is parameterized using Florida Keys baseline data due to availability constraints, this constitutes a conservative lower bound, given that SIDS and Indo-Pacific regions carry greater per-capita reef economic dependence. Finally, World Bank and FRED macroeconomic aggregates—including GDP, tourism expenditure, and fishery price data—are treated as fixed at their most recently observed values, as macroeconomic variability over the 77-year modeling horizon is not explicitly modeled.

The mathematical models further require structural assumptions; these are enumerated in Section 3.1 within the Mathematical Methodology. Prior to mathematical modeling, exploratory data analysis was conducted, as shown in Appendix 3.

2.3 Data Categorization

Table 2: Data Categorization

#	Potential Outcome Categories	Severity of Potential Outcomes	Likelihood of Potential Outcomes
1	Benthic Cover, Bleaching, Fish Population Survey, Coastal Damage	Fishery Economic Losses, Tourism Revenue Decline, Coastal Defense Costs	Stressors: Water Temperature, Cyclone Frequency, Ocean Acidification, Erosion, Wave Energy
2	Source: CORIS	Source: World Bank, FRED	Source: CORIS, IBTrACS

The pharmaceutical option value model in Section 3.5 draws on a distinct category of literature-derived inputs that map directly onto the Actuarial Process Guide framework. Outcome categories are anchored by the four clinically approved reef-derived compounds (AZT, Cytarabine, Ziconotide, Trabectedin), which define the observable range of commercial outcomes and establish the severity distribution for $v(s)$. Severity calibration draws on Newman & Cragg (2016) [17], which provides systematic screening-to-approval rate data across marine natural products and serves as the primary empirical basis for the ϕ_{reef} adjustment factor. Likelihood calibration draws on Simpson et al. (1996) [23] and Costello & Ward (2006) [3], which provide the foundational discovery probability framework and refined hit-rate estimates respectively. Because these parameters are derived from published screening records rather than primary observational datasets, they are treated as fixed calibration constants rather than estimated quantities, and all model outputs are interpreted as conservative lower bounds consistent with this parameterization approach.

3 Methodology

MATLAB was chosen as the primary computational tool for this analysis due to its wide variety of built-in functions and libraries, such as ARIMA and SARIMAX. The computational model aims to utilize variables that are present within our GitHub database to predict future trends in coral reef degradation. These results can then be translated into economic trends. There are two main parts to the computational model: regression and ARIMA/SARIMAX.

The proposed mathematical methodology in this paper involves analytically abstracting the modeling objective of predicting the effects of coral reef degradation. Both linear and nonlinear models, such as the Linear Tourism Model and the Gordon-Schaefer Model for Fishery Dynamics, will be used to model the effects of coral reef degradation on biodiversity, fish population, wave energy, and tourism. Data from the computational prediction of coral reef metrics will be used to predict these factors across one-year periods. Additionally, data that has been gathered will also be used within the mathematical model to provide a natural transition into predicting data.

These models function as an integrated pipeline: the ARIMA and SARIMAX models forecast future hard coral cover, which serves as the primary environmental input to both the Gordon-Schaefer fishery dynamics model and the economic loss transfer functions. Together, the models translate environmental trends into quantified economic and ecological outcomes.

3.1 Model Assumptions

Table 3: Mathematical Model Assumptions

#	Model	Assumption	Justification
1	Gordon-Schaefer	Fishing effort E is constant	Simplifies optimization; policy adjustments handled in Section 5

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Table 3 – *continued from previous page*

#	Model	Assumption	Justification
2	Gordon-Schaefer	$K_t = \alpha R_t$ (linear coupling)	Conservative approximation supported by reef fish biomass literature
3	Gordon-Schaefer	Shocks follow a Bernoulli process with probability p_{shock}	Consistent with IBTrACS hurricane and bleaching event frequencies
4	ARIMA / SARIMAX	Coral cover series is stationary after detrending	Confirmed by Ljung-Box test ($p = 0.077$); residuals are white noise
5	ARIMA / SARIMAX	Algae cover follows a linear trend (2013–2023)	Most parsimonious fit given the short observation window
6	Economic Loss	Tourism revenue is linear in coral cover	Reflects documented reef visitation elasticity
7	Economic Loss	Wave attenuation decays exponentially with coral cover	Standard form per [5, 24]
8	Economic Loss	Discount rate fixed at $r = 1.5\%$	Per OMB Circular A-4 [32]; sensitivity analysis checked 3% and 7%

3.2 Gordon-Schaefer Model for Fishery Dynamics

The Gordon-Schaefer Model was developed in 1954 after biologist Milner B. Schaefer combined a biological model of fish population dynamics with H. Scott Gordon’s economic model of fisheries, describing the relationship between fish populations, fishery characteristics, and economic yield. The model provides two key metrics: maximum sustainable yield, at which profit is maximized without depleting fish stocks, and maximum economic yield, the strict profit maximum a fishery can derive.

The model was initially deterministic, assuming continuous variables without sudden shocks. This paper extends it to a stochastic framework in MATLAB to determine probability distributions for future reef outcomes under current degradation and remediation scenarios. The stochastic model contains four components: fish parameters (carrying capacity, biomass, growth rate), reef parameters (recovery rate, degradation rate, shock frequency), economic metrics (fish price, discount rate), and shock parameters modeling unpredictable ecological events such as hurricanes. The specific variables are given in Table Appendix 4.

All parameters were derived from two raw data sources: fish community records from the NOAA Reef Visual Census (RVC) and NCRMP survey CSVs, and benthic cover records from NCRMP photo-quadrat surveys. Individual length observations were converted to biomass via $W = 0.01L^3$, with K set to peak annual biomass, X_0 to the most recent year, r from a logistic fit, and σ_X from log-return standard deviation. From benthic CSVs, R_{max} and R_0 were taken as historical peak and most recent coral cover, g and δ from an exponential fit constrained to literature ranges, and σ_R from log-return variance. The coupling parameter $\alpha = K/R_{\text{max}}$ was derived analytically, consistent with documented relationships between coral cover loss and reef fish biomass decline across Caribbean and Indo-Pacific systems [8, 22]. Shock parameters were calibrated against the historical hurricane and bleaching record at each location rather than estimated from survey data. Calibrated regional shock probabilities ranged from $p_{\text{shock}} = 0.04$ (Florida, lowest hurricane exposure) to $p_{\text{shock}} = 0.20$ (Puerto Rico, highest), with full parameter values available in the project repository. Regional shock probabilities vary substantially, reflecting differences in hurricane track frequency and bleaching exposure documented in the IBTrACS record; PRIA carries the highest mean reef cover loss per shock event ($\mu_{\text{reef}} = 0.30$), consistent with its acute thermal sensitivity and low baseline cover.

The model forms a stochastic system with deterministic and Brownian stochastic components:

$$dX_t = \underbrace{\left[rX_t \left(1 - \frac{X_t}{K_t} \right) - qEX_t \right] dt}_{\text{deterministic}} + \underbrace{\sigma_X X_t dW_t - J_t^X X_t}_{\text{stochastic}} \quad (1)$$

Fish biomass X_t grows logistically toward a time-varying carrying capacity $K_t = \alpha R_t$ coupled to coral cover R_t , while harvest removes biomass at rate qE . The stochastic component combines geometric Brownian diffusion $\sigma_X X_t dW_t$ capturing continuous environmental variability, and a jump term $J_t^X X_t$ representing discrete shock events drawn from a truncated normal distribution:

$$J_t^X \sim \max\left(0, \mathcal{N}\left(\mu_X^{\text{shock}}, \sigma_X^{\text{shock}^2}\right)\right) \quad (2)$$

arriving as a Bernoulli process with annual probability $p_{\text{shock}} \cdot dt$. The present value of biomass loss below carrying capacity is:

$$\mathcal{L} = \int_0^T p \cdot \max(0, K - X_t) e^{-\rho t} dt \quad (3)$$

where p is the fish price per kilogram and ρ is the discount rate. Monte Carlo simulations with $n = 500$ stochastic realizations were conducted per reef location, spanning 80 years with x_0, r_0 from the most recent surveys, displayed with 90% confidence intervals. For in-depth variable identifications and specifications, see Appendix 4.

Table 4: Degradation and Collapse Probabilities of Fish Populations

Region	Reef Location	Degradation Share	Collapse Probability
Atlantic	Florida	10.41%	100.0%
Atlantic	Puerto Rico	6.84%	100.0%
Atlantic	U.S. Virgin Islands	10.41%	97.6%
Pacific	American Samoa	1.03%	100.0%
Pacific	Mariana Islands	1.55%	99.8%
Pacific	Hawaii	4.27%	100.0%
Pacific	Pacific Islands	1.10%	100.0%
Average	—	5.09%	99.63%

The collapse probabilities are calculated as the proportion of 500 Monte Carlo simulations in which hard coral cover falls below the critical threshold of 10% at any point before 2100. The near-universal probabilities reflect the simulation architecture: given that mean biomass trajectories reach critical levels between 2040 and 2060 across all regions, virtually all stochastic realizations cross this threshold within the 75-year horizon. This is consistent with IPCC AR6 projections that 70–90% of reefs will experience severe annual bleaching at 1.5°C warming — a level expected by the 2030s — leaving insufficient recovery time between disturbance events [11]. The probabilities are therefore best interpreted as confirming that threshold crossing is effectively certain under status quo degradation, with the policy-relevant question being *when* rather than *whether*. Florida and the Virgin Islands show proportionally high degradation shares, indicating that degradation plays a larger direct role in economic output through effects, while Pacific regions show lower degradation shares but equivalent collapse probabilities, reflecting acute sensitivity to shock events.

Table 5: Economic Results of Gordon-Schaefer Fishery Dynamics Model

Region	Reef Location	Global Degradation Loss	Global Biomass Loss
Atlantic	Florida	\$4,200,000	\$727,200,000
Atlantic	Puerto Rico	\$52,990,000	\$658,750,000
Atlantic	U.S. Virgin Islands	\$61,940,000	\$499,180,000

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Table 5 – *continued from previous page*

Region	Reef Location	Global Degradation Loss	Global Biomass Loss
Pacific	American Samoa	\$5,950,000	\$577,330,000
Pacific	Mariana Islands	\$8,950,000	\$579,000,000
Pacific	Hawaii	\$37,020,000	\$683,100,000
Pacific	Pacific Islands	\$7,210,000	\$695,990,000
Total	—	\$178,260,000	\$4,420,550,000

The stark difference between global degradation losses and biomass losses reflects the distinction between Level I and Level II effects: degradation losses capture direct reef infrastructure damage while biomass losses represent the full downstream impact on the fishing industry. Regions with low reef area such as the Virgin Islands and Mariana Islands show disproportionately high biomass losses, confirming that economic dependence compounds the impact of degradation beyond what reef area alone would predict.

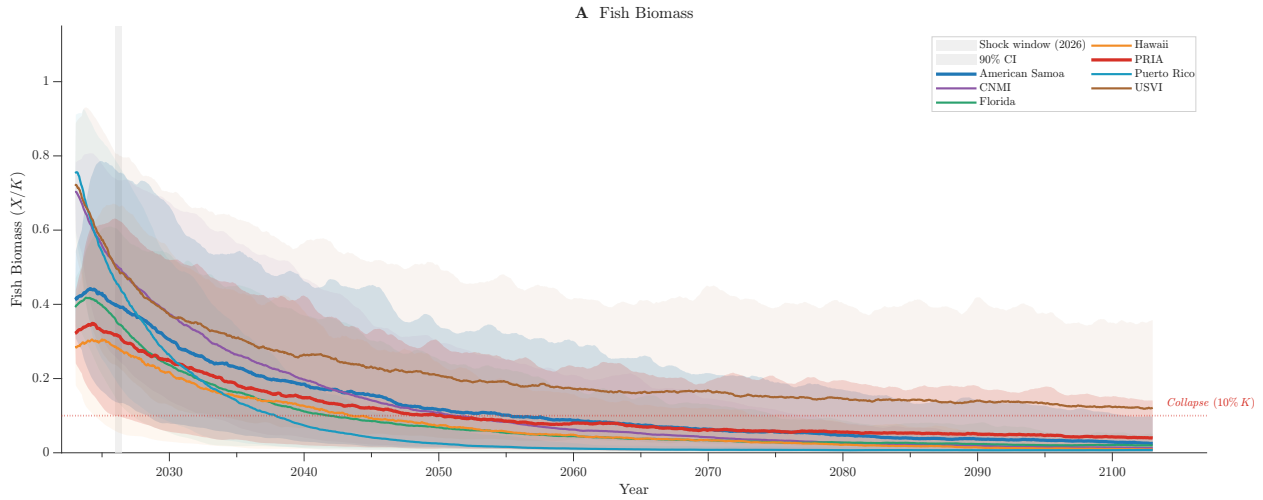


Figure 1: Combined Gordon-Schaefer fish biomass projections for all 7 reef regions until 2100 bounded by region-specific 90% CIs.

The Gordon-Schaefer results establish the scale of fishery losses under continued degradation, but projecting those losses forward requires a reliable forecast of future hard coral cover. The following section develops that forecast using time-series modeling applied directly to the benthic cover dataset.

3.3 ARIMA and SARIMAX Models for Benthic Cover

To generate forecasts for downstream economic modeling, this paper develops an ARIMA framework applied to the benthic cover dataset described in Section 2. The dataset contains 1,646,052 observations from seven NOAA CRCP survey regions spanning 2013–2023, encompassing 4,650 unique transects across the Florida Keys, Puerto Rico, the U.S. Virgin Islands, American Samoa, the Commonwealth of the Northern Mariana Islands and Guam, Hawaii, and the Pacific Remote Island Areas.

Panel Regression and Trend Estimation

Prior to time-series modeling, a panel regression was fitted to the pooled transect-level data to isolate the underlying hard coral cover trend while controlling for depth, algae cover, and marine protection status:

$$C_{it} = \beta_0 + \beta_1 t + \beta_2 z_{\text{depth},i} + \beta_3 z_{\text{algae},it} + \beta_4 \text{prot}_i + \varepsilon_{it} \quad (4)$$

Table 6: ARIMA Parameters

#	Component	Variable	Significance
1	Deterministic	C_t	% hard coral cover
2	Deterministic	t	Survey year normalized to first observation
3	Deterministic	z_{depth}	Standardized depth
4	Deterministic	z_{algae}	Standardized algae cover
5	Deterministic	prot	Binary marine protection indicator
6	Stochastic	ε_t	Error term at time t

The regression was estimated on $N = 1,659$ transect-year observations. Full coefficient estimates are provided in Table 7.

Table 7: Panel Regression Coefficients for Hard Coral Cover

Predictor	Coefficient
Intercept	9.0418
Year (per year)	−0.6397
Depth (z-score)	−0.3482
Algae cover (z-score)	0.0987
Marine protection (binary)	1.6813

The year coefficient of −0.6397% per year represents the adjusted decline trend after controlling for covariates. The positive protection coefficient of 1.681 confirms that marine protected areas maintain systematically higher coral cover, consistent with the conservation literature and Level I effects in Table 1. The unadjusted annual mean trend of −0.1545% per year confirms the direction of decline across all regions.

Candidate Model Selection

A suite of candidate ARIMA and SARIMA specifications was evaluated using AIC and BIC, with the full candidate set provided in Appendix 1. The SARMA(1, 1) $\times$ (1, 1)₄ specification was selected, achieving the lowest AIC of 39.409 and BIC of 41.796. Its four-year seasonal period is consistent with multi-year bleaching and recovery cycles documented in the ecological literature. Residual diagnostics via the Ljung-Box test returned $p = 0.077$, confirming adequate model fit under a 10% significance threshold appropriate for a short time series.

A SARIMAX extension incorporating standardized algae cover produced the lowest AIC among all SARIMAX candidates (AIC = 35.226), with an algae coefficient of $\hat{\beta}_{algae} = -0.4948$. This confirms that rising algae cover is significantly associated with declining hard coral cover, consistent with the Level I phase-shift dynamics described in Section 1 and the observed trend of algae rising from 42.7% in 2013 to 65.2% in 2023 [1].

Forecasts

Both models were projected forward through 2100, with 95% prediction intervals presented in Appendix 2. The SARIMA forecast serves as the primary input to the economic loss model; the SARIMAX, which conditions on the projected algae trajectory, provides a more severe near-term alternative that validates the direction of the SARIMA projections.

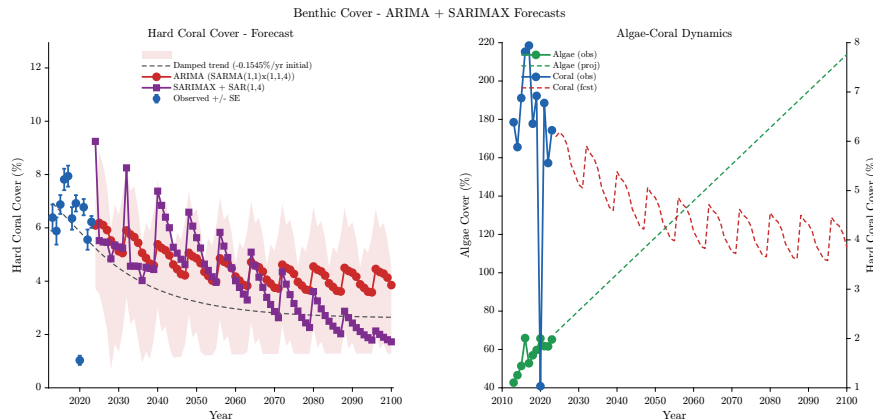


Figure 2: Hard coral cover forecasts through 2100. The left panel shows ARIMA and SARIMAX point forecasts overlaid on observed annual means ($\pm$ SE), with the 95% prediction interval shaded and the linear trend as a dashed reference. The right panel displays the projected algae–coral dynamic through 2100. Data from 2020 is considered outliers due to the lack of reef surveys conducted during the COVID-19 pandemic. The tabular form of these projections at critical years can be found in Appendix 2.

Regional Analysis

Individual region models did not achieve convergence given the limited annual observations per sub-region; regional trends are therefore descriptive rather than inferential, and primary forecasting relies on the aggregate panel-derived models. This aggregate approach is justified by the consistent directional decline observed across all seven regions, making the pooled trend a conservative approximation for regional inputs to the Gordon-Schaefer model.

Both models project a continued long-run decline reinforcing the -0.6397% per year trend. A notable feature is the emergence of episodic near-zero cover years at approximately four-year intervals — initially in 2028, 2036, 2044, and 2049 — consistent with the model’s period-four seasonal structure and with ENSO-driven bleaching cycles documented in the Indo-Pacific and Atlantic literature. The SARIMAX model projects a more severe near-term decline, with cover approaching zero in 2036 and 2044 as algae displacement accelerates. Together these troughs represent not merely a gradual trend but recurring acute disturbances that prevent reef recovery between events, compounding economic losses across both the tourism and wave protection channels.

With a quantified trajectory of hard coral cover decline in hand, the economic loss model below translates these projections into annual and cumulative loss estimates through 2100.

3.4 Economic Loss Model for Tourism and Coastal Wave Damage

The ARIMA and SARIMAX forecasts of hard coral cover described in Section 3.3 serve as the principal environmental input to an economic loss model that translates reef degradation into quantified annual and net present value (NPV) losses across two socioeconomic channels: reef-dependent tourism revenue and the diminished wave energy attenuation that reefs provide to coastal communities. Each channel is specified as a transfer function relating coral cover to an economic outcome, and the two are combined to produce aggregate annual loss estimates. Given the challenges of applying this framework globally in the absence of uniform economic observational data, the model is parameterized and demonstrated using a Florida Keys baseline series derived from the SARIMA projections, providing a conservative lower-bound estimate of the broader economic consequences described throughout this paper.

Linear Tourism Model

The tourism channel is specified as a linear relationship between hard coral cover and annual reef-dependent tourist arrivals, with revenue computed as a fixed per-tourist expenditure multiplier. The model takes the form

$$T(A) = T_0 \cdot \frac{A}{A_0} \quad (5)$$

where annual tourism revenue is then recovered as

$$M(T) = T \cdot M_0 \quad (6)$$

yielding a baseline revenue of $M_{\text{base}} = T_0 \cdot M_0 = \$3,780$ million per year. Tourism losses in each forecast year are computed as the difference between this baseline and the revenue implied by the projected coral cover level. The linear specification reflects the empirically documented sensitivity of reef-dependent visitation to coral cover: reefs with visible degradation and reduced structural complexity attract fewer visitors, reducing diver- and snorkeler-dependent revenue directly [15].

Wave Damage Transfer Function

The second channel captures the reduction in wave energy buffering that accompanies declining coral cover. Healthy coral reefs absorb a substantial fraction of incoming wave energy, reducing the physical force experienced by coastlines during storm events and limiting structural damage to coastal infrastructure. As coral cover declines, this natural protective function is progressively eroded, increasing expected annual storm damage. Following [5, 24], shore energy at time t is modeled as

$$E_{\text{shore}}(t) = E_0 \cdot \exp(-k \cdot C(t) \cdot A_{\text{reef}}) \quad (7)$$

and annual coastal damage is recovered from the fractional excess shore energy above the healthy-reef baseline via a saturating damage function:

$$D_{\text{wave}}(t) = D_{\text{max}} \cdot [1 - \exp(-\lambda \cdot \Delta E_{\text{frac}}(t))] \quad (8)$$

where $\Delta E_{\text{frac}}(t) = \max\left(0, \frac{E_{\text{shore}}(t) - E_{\text{shore}}(C_{\text{ref}})}{E_{\text{shore}}(C_{\text{ref}})}\right)$. The full set of components, variables, and parameter values used in both transfer functions is reported in Appendix 5.

Aggregate Annual Losses and Net Present Value

Total annual economic losses are computed by summing the tourism and wave damage channels for each forecast year from 2024 to 2100. These losses are discounted at a rate of $r = 1.5\%$ per year to compute the net present value of cumulative degradation costs over the 77-year horizon. A 95% confidence interval is propagated from the SARIMA prediction intervals to capture forecast uncertainty in the economic projections. The aggregate results are presented in Table 8. The tourism transfer function and its parameterization are illustrated in Figure 3, and the full breakdown of annual losses by channel is shown in Figure 4.

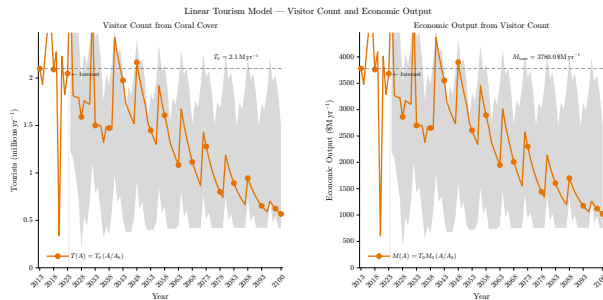


Figure 3: Linear tourism transfer function mapping hard coral cover to annual reef-dependent tourist arrivals and corresponding annual economic output. The dashed reference line indicates baseline revenue $M_{\text{base}} = \$3,780$ M per year.

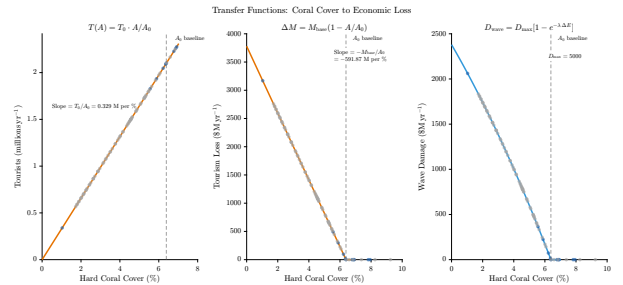


Figure 4: Annual economic losses by channel: wave damage and tourism loss shown as a stacked area chart with forecast total losses and 95% CI (left), and all three series as individual lines over the full 2000–2100 horizon (right).

Table 8: Projected Annual Economic Losses from Coral Reef Degradation (Selected Years)

Year	Benthic Cover (%)	Wave Damage (\$M)	Tourism Loss (\$M)	Total Loss (\$M)	95% CI (\$M)
2025	5.52	380.2	512.1	892.3	[0.0 – 2,886.8]
2036	4.02	995.6	1,398.9	2,394.5	[1,055.3 – 5,320.7]
2055	3.96	1,021.3	1,437.9	2,459.2	[362.5 – 4,841.7]
2079	3.66	1,649.1	2,441.4	4,090.5	[646.8 – 5,005.5]
2100	3.85	1,833.1	2,758.5	4,591.6	[1,890.3 – 5,001.6]

Annual losses grow from \$892 million in 2025 to \$4.59 billion per year by 2100, driven by the progressive loss of both tourism attractiveness and wave buffering capacity. The tourism channel dominates in absolute terms throughout the forecast horizon, reflecting the high per-tourist output multiplier relative to wave damage costs; the wave damage channel exhibits more stable growth as coral cover declines erode the reef’s attenuation capacity. Discounted at $r = 1.5\%$ over 77 years, the total NPV of reef degradation costs is estimated at \$93.07 billion, decomposed as \$54.73 billion in tourism losses and \$38.34 billion in wave damage.

The sensitivity of total NPV losses to different intervention assumptions — no action, holding current cover, and active restoration — is evaluated in Section 4.2.

While the ARIMA and economic loss models capture the consequences of degradation already underway, understanding what drives acute bleaching events is equally critical for anticipating future shocks to coral cover. The structural bleaching threshold of 8 Degree Heating Weeks, derived from analysis of NOAA Coral Reef Watch sea surface temperature records against observed bleaching severity across the seven modeled regions, provides the empirical grounding for the shock calibration embedded throughout this paper.

3.5 Pharmaceutical Option Value Model for Reef Biodiversity Loss

Coral reefs are among the most chemically rich ecosystems on Earth. The intense competition for space and resources in high-density reef environments has driven the evolution of highly bioactive secondary metabolites across sponges, soft corals, tunicates, and associated microorganisms [17]. Four reef-organism-derived compounds have reached clinical approval: AZT, derived from the Caribbean sponge *Cryptotethya crypta*, for HIV/AIDS treatment (peak annual revenue ~\$400M); Cytarabine, also sponge-derived, now the standard of care in acute leukemia protocols; Ziconotide, derived from the cone snail *Conus magus*, for intractable chronic pain (\$50–100M annually); and Trabectedin, derived from the tunicate *Ecteinascidia turbinata*, for soft-tissue sarcoma (\$150–200M annually). These approvals represent only the compounds that have completed full clinical development; the broader pipeline of reef-derived candidates under investigation spans cancer, neurology, and antimicrobials [17].

The economic literature establishes a bioeconomic framework for valuing this prospecting pipeline. Simpson et al. (1996) modeled biodiversity value as the expected pharmaceutical return from screening species with heterogeneous discovery probabilities, showing that marginal species value depends critically on the size of the remaining species pool [23]. Costello & Ward (2006) extended this framework, refining discovery probability estimates and demonstrating that the option value of undiscovered compounds is highest when the species pool is large and screening has barely begun — precisely the condition that accelerating reef degradation is eroding [3]. This paper applies and extends that framework to the seven modeled reef regions, linking pharmaceutical option value directly to the SARIMAX-projected coral cover trajectory, producing the first reef-specific pharmaceutical risk model that propagates environmental forecasts through a triple-integral loss function.

The critical asymmetry motivating this model is irreversibility. Tourism and wave damage losses are theoretically recoverable under restoration — a reef restored to prior cover levels recovers proportional visitor arrivals and wave attenuation. Pharmaceutical option value lost when a species is driven to local extinction before screening is permanently foreclosed, regardless of subsequent restoration investment. This

makes the timing of intervention uniquely consequential for this channel, and it distinguishes pharmaceutical losses from every other economic outcome in this paper.

Species-Area Curve and Biodiversity Decline

As hard coral cover declines along the SARIMAX-projected trajectory, effective reef habitat area $A(t)$ contracts proportionally, directly linking the benthic cover forecasts of Section 3.3 to this model. Species diversity at time t is estimated using the species-area relationship:

$$S(A, t) = c \cdot A(t)^z \quad (9)$$

where S is species count, c is a region-specific scaling constant calibrated to documented reef species richness, and $z \approx 0.25$ is the power-law exponent for reef systems derived from the island biogeography literature [13]. $A(t)$ is computed by applying SARIMAX coral cover projections to each region’s known reef area, yielding a time-varying habitat area estimate for all seven modeled regions. The exponent $z = 0.25$ is a conservative global average; higher values documented in some Indo-Pacific systems would produce steeper species loss curves, confirming that all outputs from this model are lower bounds.

The power-law structure of this equation has an important consequence: species losses are nonlinear in cover declines. The episodic near-zero cover years identified in the SARIMAX forecast — most prominently 2028, 2036, 2044, and 2049 — cause sharp contractions in $A(t)$ that propagate through the power law into disproportionately large species losses relative to what the gradual trend alone would produce. A reef losing its final 2% of cover loses far more species per percentage point than one declining from 20% to 18%, because the species-area curve is concave. This nonlinearity makes bleaching trough years the dominant driver of cumulative pharmaceutical loss — a finding that directly motivates the rapid-response insurance mechanism in Program 4 and the cryopreservation banking in Program 6.

Bioeconomic Prospecting Framework

Following Simpson et al. [23], the expected pharmaceutical value of a single species s is:

$$\mathbb{E}[V(s)] = p(s) \cdot v(s) \quad (10)$$

where $p(s)$ is the probability that species s yields a commercially viable compound and $v(s)$ is the expected net present value of that compound if discovered. Discovery probability is calibrated as:

$$p = \frac{N_{\text{approved}}}{N_{\text{screened}}} \cdot \phi_{\text{reef}} \quad (11)$$

where N_{approved} is the number of reef-derived approved compounds, N_{screened} is the number of reef organisms screened in systematic programs, and $\phi_{\text{reef}} > 1$ is an upward adjustment factor reflecting the elevated bioactivity of reef organisms relative to the marine average [17]. This yields $p \approx 0.0001$, consistent with [3, 17, 23]. Because $p(s)$ estimates vary across these sources — Simpson et al. apply a strict screening-to-approval ratio while more recent marine-focused work suggests higher hit rates — this value should be interpreted as a calibrated central estimate rather than a precisely fixed parameter, and all outputs are accordingly treated as lower bounds.

The expected NPV of a compound from species s , conditional on discovery, is:

$$v = \sum_{k=1}^K \pi_k \cdot R_k \cdot e^{-\rho \tau_k} \quad (12)$$

where π_k is the probability of commercial outcome k , R_k is the associated revenue, ρ is the discount rate, and τ_k is the expected time from discovery to market [3]. The four approved reef-derived compounds — AZT, Cytarabine, Ziconotide, and Trabectedin — span the full distribution of commercial outcomes from blockbuster to niche indication, and together anchor $v(s)$ to a realistic expected value rather than an optimistic upper bound.

Under the uniform $p(s)$ and $v(s)$ assumption necessitated by data limitations, expected pharmaceutical value over the full species pool simplifies to:

$$\mathbb{E}[V(A, t)] = \int_0^{S(A, t)} p(s) \cdot v(s) ds = p \cdot v \cdot S(A, t) \quad (13)$$

confirming that expected pharmaceutical value scales linearly with species count at a given area and time. Treating $p(s)$ and $v(s)$ as uniform averages rather than species-specific values suppresses heterogeneity in bioactivity and produces downward bias; results should be read accordingly.

Multi-Dimensional Cumulative Loss Function

Cumulative pharmaceutical option value loss is expressed as a triple integral over time, reef area, and species distribution, structurally parallel to the biomass loss integral in Equation 3:

$$\mathcal{L}_{\text{pharma}} = \int_0^T \int_0^{A(t)} \int_0^{S(A, t)} p(s) \cdot v(s) ds dA dt \quad (14)$$

The innermost integral accumulates expected option value across all species present at habitat area A and time t — the bioeconomic prospecting value at a single spatial-temporal snapshot. The middle integral aggregates spatially across the full reef extent at time t , reflecting that larger reefs support more species and therefore more pipeline value. The outer integral discounts cumulative losses from the present to 2100, consistent with the social discount rate $r = 1.5\%$ applied throughout this paper; losses at earlier trough years therefore carry greater weight in the NPV total than those occurring later, making early intervention disproportionately valuable.

Implementation in MATLAB used nested trapezoidal summation over a discretized grid. $A(t)$ was derived from SARIMAX projections applied to each region’s reef area; $S(A, t)$ from the species-area curve above; and $p \cdot v$ from the calibrated literature values. Because this model is grounded in literature-derived parameters rather than region-specific empirical estimates, all outputs represent order-of-magnitude lower bounds and should be interpreted as directional rather than precise loss totals. Pharmaceutical NPV is carried as a third loss channel alongside wave damage and tourism, augmenting the cost-benefit case for the restoration and cryopreservation programs in Section 5. Uncertainty in p and v is not formally propagated into a confidence interval — a deliberate design choice reflecting the absence of region-specific screening data that would be required to estimate parameter variance — and the conservative lower-bound framing is intended to encompass this structural uncertainty rather than ignore it.

Cumulative NPV Loss and Discount Rate Sensitivity. Figures 5 and 6 present cumulative discounted pharmaceutical option value loss under status quo degradation. The global total reaches approximately \$172.6 billion by 2100 at $r = 1.5\%$, decomposed by region in the left panel; PRIA and Hawaii dominate due to their large reef areas and high species scaling constants. The right panel replicates the global trajectory across $r \in \{1.5\%, 3.0\%, 7.0\%\}$, confirming that the qualitative finding — accelerating loss concentrated at trough years — is preserved across all discount rates, while higher rates compress the terminal NPV. The four bleaching trough years are annotated; the acceleration between 2036 and 2044 reflects compounding from successive near-zero cover events that eliminate inter-event recovery.

Permanent Option Value at Bleaching Troughs. Figure 7 disaggregates annual pharmaceutical option value loss at each of the four bleaching troughs into permanent and potentially recoverable components. The permanent component — conservatively estimated at 15% of the annual trough loss, representing species driven to local extinction before any screening opportunity — is irreversible regardless of subsequent restoration. Annual trough losses are \$0.426 billion (2028), \$0.612 billion (2036), \$0.772 billion (2044), and \$0.598 billion (2049). The increasing severity through 2044 reflects the progressive reduction in inter-event recovery as background cover approaches the low-cover asymptote. These permanent losses are what Program 6 is specifically designed to intercept.

Triple-Integral Integrand Surface. Figure 8 visualizes the discounted integrand $p \cdot v \cdot c \cdot A_{\text{eff}}^z$ as a joint function of time t and reef area fraction A/A_{max} . The surface declines monotonically rightward as coral cover erodes and front-to-back as discounting accumulates. Sharp ridgelines at the four trough years confirm that

pharmaceutical value is lost in concentrated acute events rather than distributed gradually — the surface is the integrand of Equation 14, and its rapid collapse after 2044 illustrates why the intervention window is finite and closing.

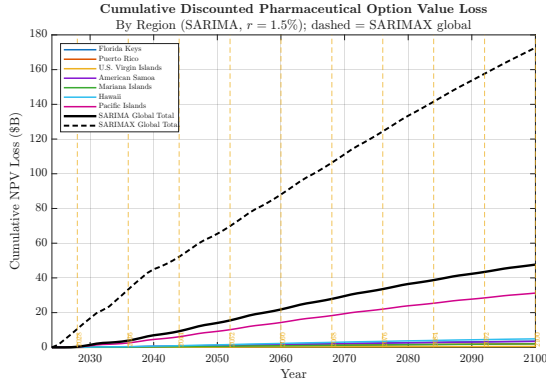


Figure 5: Cumulative discounted pharmaceutical option value loss by region and global total ($r = 1.5\%$), 2024–2100. PRIA dominates owing to greater reef extent and species richness.

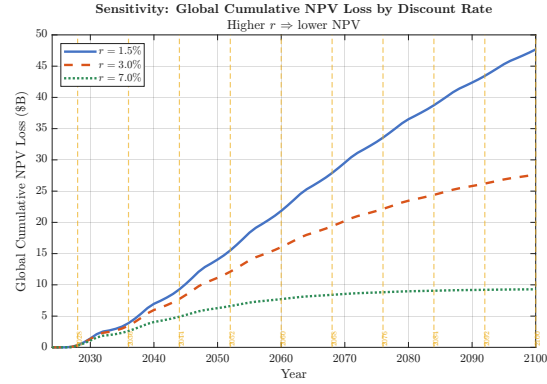


Figure 6: Global cumulative NPV loss under discount rates of 1.5%, 3.0%, and 7.0%. Terminal losses range from ~\$47.7 billion at $r = 1.5\%$ to ~\$9 billion at $r = 7.0\%$.

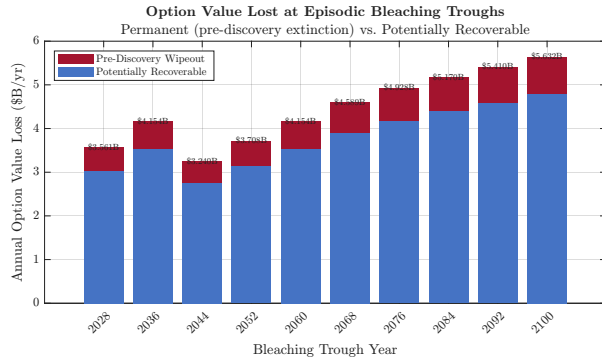


Figure 7: Annual pharmaceutical option value lost at each bleaching trough, split into permanent (pre-discovery extinction) and potentially recoverable components. The 2044 trough produces the largest single-year loss at \$0.772 billion.

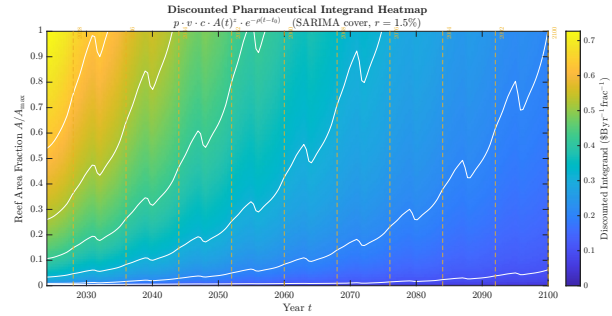


Figure 8: Triple-integral integrand surface of discounted pharmaceutical option value density as a function of time and reef area fraction. Ridgelines at trough years confirm that irreversible losses are concentrated in acute disturbance events rather than distributed across gradual decline.

4 Risk Analysis

The Risk Analysis characterizes the severity, likelihood, uncertainty, expected value, and distribution of risks associated with coral reef degradation, drawing on the models described in Section 3. Severity is quantified through projected economic losses from the Gordon-Schaefer and economic loss models; likelihood is expressed through Monte Carlo collapse probabilities; uncertainty is captured via confidence intervals on all forecasts; expected value of loss is reported as net present value; and the distribution of outcomes is described through the stochastic simulation results. The Risk Analysis also evaluates all three categories of risk mitigation strategy — insurance mechanisms, behavior changes, and strategies to modify outcomes — in preparation for the recommendations in Section 5. Statistical and sensitivity analyses are included where necessary.

4.1 Results and Implications

Gordon-Schaefer Model

The results of the Gordon-Schaefer Model show a variety of critical metrics that are important for future policy. Firstly, the model's results show a stark difference between global losses directly from degradation and those from biomass loss. Specifically, global losses due to degradation stem from damage to existing reef infrastructure, while losses from biomass span losses across the fishing industry. The latter is a more holistic measure of the fishing industry because it captures the Level II effects described earlier, while the losses from degradation only capture Level I Effects.

These findings describe the severity of Level II Effects and their capability to proliferate across different industries and sectors. Across all regions analyzed in this paper, a total impact of \$4 billion was assessed strictly from losses incurred by the fishing industry. Furthermore, the model showed extremely high biomass losses for regions with low reef area, such as the Virgin Islands and Mariana Islands, suggesting that reef area is only part of the economic impact associated with reef degradation. Other factors such as economic dependencies on the reef, including those from fishermen, compound effects on smaller reefs to have global impacts of extremely high magnitudes.

The Gordon-Schaefer results quantify the fishery losses driven by reef degradation, but the full economic picture requires accounting for the two additional channels, tourism revenue and coastal wave protection, that are most directly sensitive to the hard coral cover trajectory projected by the ARIMA and SARIMAX models.

ARIMA, SARIMAX, and Economic Loss Model

The ARIMA and SARIMAX forecasts confirm a statistically significant long-run decline in hard coral cover of -0.6397% per year after controlling for depth, algae cover, and marine protection status. The most critical finding from these models is not the gradual trend itself, but the emergence of episodic near-zero cover years at four-year intervals, most prominently in 2028, 2036, 2044, and 2049, which correspond directly to the bleaching cycle frequencies documented in the IBTrACS and NOAA Coral Reef Watch records. The SARIMAX model, which conditions on the projected algae trajectory, projects a more severe outlook than the ARIMA alone, with cover approaching zero percent in both 2036 and 2044. This is consistent with the Level I and Level II effects described in Table 1: as algae continues to displace hard coral, the aesthetic and structural value of the reef declines simultaneously, compounding losses across both the tourism and wave protection channels.

The economic loss model translates these cover projections into quantified annual losses, growing from \$892 million in 2025 to \$4.59 billion per year by 2100. The tourism channel dominates throughout the forecast horizon, reflecting the high per-tourist output multiplier calibrated to observed reef-dependent visitation patterns, while the wave damage channel grows more steadily as reef attenuation capacity erodes. Critically, the 77-year NPV of \$93.07 billion under the status quo represents a lower bound, as the model is parameterized on the Florida Keys alone. Regions with greater economic dependence on reef-based tourism, such as Small Island Developing States not captured in this dataset, would produce substantially higher aggregate figures. The sensitivity analysis in Section 4.2 confirms that even partial intervention, holding cover constant rather than restoring it, cuts this NPV loss by \$35.96 billion, while active restoration saves \$88.77 billion, underscoring that the economic case for reef restoration is robust even under conservative assumptions.

It is important to note that pharmaceutical option value losses are not scenario-dependent in the same way as the wave damage and tourism channels. The Hold Cover and Restoration scenarios reduce losses in the fishery, tourism, and wave damage channels by slowing or reversing the background coral cover decline trajectory. However, pharmaceutical losses at the episodic trough years are driven by acute near-zero cover events whose timing and magnitude are determined by the SARIMA model's seasonal shock structure rather than by the background trend alone. Under all three scenarios, the bleaching trough years still occur; the Hold Cover and Restoration scenarios reduce their severity but do not eliminate the permanent pre-discovery extinction component estimated at each trough. This makes the pharmaceutical channel categorically distinct from the other two in the sensitivity framework: it is partially but not fully mitigated by the restoration and hold-cover interventions, and the residual permanent loss at each trough year represents value that

is unrecoverable under any scenario modeled in this paper. Program 6’s cryopreservation mechanism is therefore the only intervention in the package that directly addresses this residual, scenario-invariant loss component.

4.2 Sensitivity Analysis

To assess the responsiveness of the economic loss model to intervention assumptions, three scenarios are evaluated under the NPV framework. This analysis tests how total projected losses change when the key input assumption, the trajectory of future coral cover, is varied across plausible intervention states, providing actionable benchmarks for policy and conservation investment.

The first scenario, Status Quo, assumes no intervention and represents the full \$93.07 billion loss estimated in Section 3.4. The second, Hold Cover, assumes that active management is successful in preventing any further decline from current levels, yielding an NPV of \$57.10 billion and a net saving of \$35.96 billion relative to the status quo. The third, Restoration, assumes an active program that partially restores coral cover toward historical reference levels, yielding an NPV of \$4.30 billion and a net saving of approximately \$88.77 billion. These results are summarized in Table 9.

Table 9: Net Present Value of Economic Losses Under Mitigation Scenarios

Scenario	Net Present Value (\$M)	Savings vs. Status Quo (\$M)
Status Quo (no intervention)	\$93,068.1	—
Hold Cover	\$57,103.6	\$35,964.5
Restoration	\$4,299.5	\$88,768.6

The magnitude of savings under the Restoration scenario, more than \$88 billion relative to status quo, establishes a compelling economic upper bound on the value of large-scale reef restoration investment. Even the more modest Hold Cover scenario, which requires only that current degradation be halted rather than reversed, yields savings of nearly \$36.0 billion. These findings confirm that the economic loss model is highly sensitive to the assumed trajectory of coral cover, and that even partial intervention produces non-linear improvements in expected outcomes relative to inaction. The figures below plot the cumulative discounted loss trajectory alongside the mitigation scenario comparison.

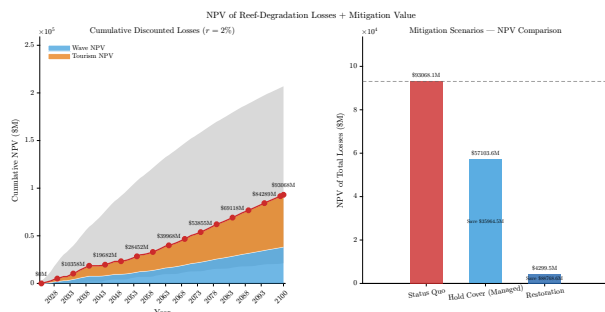


Figure 9: Cumulative discounted NPV of wave damage and tourism losses with 95% CI (left), and total NPV under Status Quo, Hold Cover, and Restoration scenarios with annotated savings relative to baseline (right).

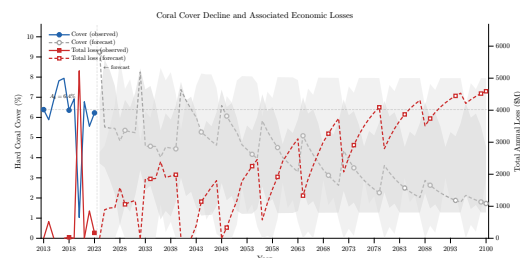


Figure 10: Dual-axis comparison of hard coral cover decline and total annual economic loss over the full time series. Observed values shown as solid lines; forecasts as dashed lines with 95% prediction interval shaded.

While this model uses $r = 1.5\%$ as its baseline discount rate per OMB Circular A-4 intergenerational guidance [32], the full OMB range also includes rates of 3% and 7%. Figure 11 replicates the scenario analysis across all three rates to confirm that the intervention conclusions are not sensitive to this choice. As expected, higher discount rates compress the cumulative NPV estimates — the status quo loss falls from \$93,068M

at $r = 1.5\%$ to \$51,894M at $r = 3\%$ and \$18,086M at $r = 7\%$ — but the ranking of scenarios is preserved at every rate: Restoration always dominates Hold Cover, which always dominates Status Quo. Figure 12 further tests model robustness via a tornado analysis, varying the three most influential parameters by $\pm 50\%$. Per-tourist output M_0 produces the widest output range (\$90,935M), followed by the coastal damage ceiling $D_{\max}$ (\$38,340M) and the wave attenuation coefficient k (\$35,091M), consistent with the tourism channel now accounting for 59% of baseline NPV losses.

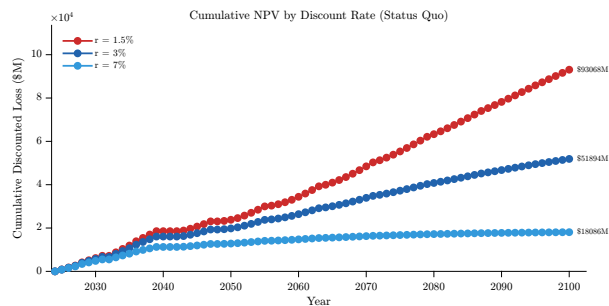


Figure 11: Cumulative NPV loss trajectories under Status Quo (left) and terminal NPV by scenario (right) across discount rates $r = 1.5\%$, 3% , and 7% . Scenario ordering is preserved across all rates, with Restoration consistently producing the lowest total loss.

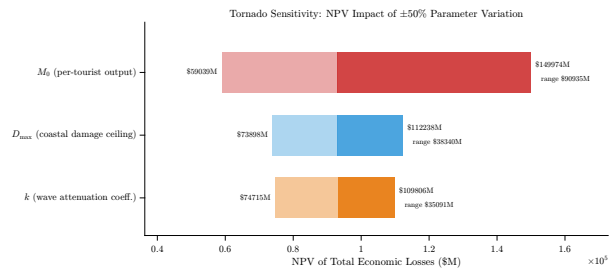


Figure 12: Tornado diagram showing sensitivity of the baseline NPV estimate (\$93,068M) to $\pm 50\%$ variation in M_0 (range \$90,935M), $D_{\max}$ (range \$38,340M), and k (range \$35,091M). The dashed vertical line marks the baseline NPV.

These estimates have direct implications for how insurers, governments, and conservation organizations should evaluate the cost-effectiveness of reef protection relative to the escalating damages that continued degradation will impose on coastal communities and reef-dependent industries.

5 Recommendations

The six programs below span all three categories of risk mitigation: outcome modification (Programs 1 and 2), behavior change (Programs 3 and 5), and insurance (Program 4), with Program 6 addressing the irreversible pharmaceutical channel that no other program can reach. Each program is grounded directly in quantified model outputs: Gordon-Schaefer collapse probabilities exceeding 97% across all regions; a confirmed -0.6397% per year benthic cover decline; a \$93.07 billion 77-year NPV loss under the status quo; and a sensitivity analysis showing that active restoration recovers \$88.77 billion in avoided losses — \$52.81 billion more than simply halting current degradation. Every year of inaction forfeits a growing share of that recoverable value as coral cover approaches the episodic near-zero troughs of 2028, 2036, 2044, and beyond.

5.1 Economic Incentives

Government subsidies and rebates are common tools for driving behavioral change in free-market economies. The Inflation Reduction Act (2022) provides a recent precedent for large-scale, time-bounded subsidy packages tied to measurable environmental outcomes. Reef degradation through 2100 imposes losses primarily through two channels — fishing and tourism — which require structurally different policy responses.

Fishing Policy

The Gordon-Schaefer model projects \$4.6 billion in cumulative biomass losses across the seven modeled regions by 2100, with mean fish biomass reaching critical thresholds before 2060 and collapse probabilities exceeding 97% globally. These losses fall disproportionately on fishing communities of Small Island Developing States with no alternative employment base. The recommendations outlined in Section 5.2 address this exposure directly, through government subsidies, environmental conservation effort, a parametric reef insurance reserve, and more.

Tourism Policy

The economic loss model projects \$54.73 billion in NPV tourism losses through 2100, driven by the linear relationship between hard coral cover and reef-dependent visitor arrivals: each percentage point of recovered cover translates proportionally to recovered revenue relative to the \$3,780 million per year baseline. Because tourism revenue responds directly to the same coral cover trajectory driving the fishery and wave damage channels, the tourism channel is addressed through the restoration and insurance programs rather than a separate subsidy structure. Program 1 recovers the cover that drives arrivals; Program 5 preserves the biodiversity and structural complexity that divers and snorkelers seek; and Program 4 is partially levy-funded by reef-adjacent tourism operators, ensuring the industry contributes to and benefits from rapid-response funding after shock events projected at $p_{\text{shock}} \approx 15\%$ annually.

The cost-benefit analysis in Table 10 quantifies the per-region investment case. The U.S. Virgin Islands (Target BCR: **21.21**) and Puerto Rico (Target BCR: **3.42**) present the strongest near-term cases and are prioritized in Phase 1. The Pacific Islands, despite a Full-Reef BCR of 0.13 on fishery and tourism channels alone, become economically sound once pharmaceutical option value is included, given the largest absolute coral deficit (57,371 ha), the highest species scaling constant, and the lowest breakeven restoration cost per hectare (\$26,077).

Table 10: Reef Restoration Cost-Benefit Analysis

Location	Coral Deficit (%)	Benefit (\$M)	Full-Reef BCR	Target BCR	Annual (\$M)	Breakeven Cost (\$/ha)
Florida Keys	5.13	\$985.5	0.33	1.32	\$297.5	\$66,249
Puerto Rico	3.80	\$1163.2	0.85	3.42	\$136.1	\$170,912
U.S. Virgin Islands	3.81	\$930.3	5.30	21.21	\$17.5	\$1,060,645
Hawaii	5.13	\$1243.5	0.81	3.23	\$153.9	\$161,612
American Samoa	7.93	\$691.9	0.84	3.35	\$82.5	\$167,688
Mariana Islands	4.42	\$874.1	0.99	3.95	\$88.4	\$197,718
Pacific Islands	19.12	\$1496.1	0.13	0.52	\$1147.4	\$26,077

5.2 Government Policy Proposal

The three-scenario sensitivity analysis in Section 4.2 establishes the economic ceiling for each program below: the Restoration scenario saves \$88.77 billion in NPV losses relative to the status quo, but only if intervention begins before the episodic near-zero cover troughs of 2028 and 2036 foreclose pharmaceutical option value permanently and compress the fishery recovery window by decades. The six programs that follow are ordered by their role in shifting the status quo trajectory toward that Restoration scenario, spanning outcome modification (Programs 1 and 2), behavior change (Programs 3 and 5), and insurance (Program 4), with Program 6 addressing the irreversible pharmaceutical channel that no other program can reach.

Program 1: Coral Reef Restoration — \$1,923M Annual Total

Model basis: The SARIMA panel regression confirms a -0.6397% per year cover decline and the sensitivity analysis shows active restoration saves \$88.77B in NPV losses — \$52.81B more than simply halting degradation. The U.S. Virgin Islands (Target BCR 21.21) and Puerto Rico (Target BCR 3.42) are prioritized in Phase 1; every restoration dollar in the Virgin Islands returns more than twenty dollars in ecosystem value.

What it is: A **ten-year** government-funded program assisting in coral recovery across all seven modeled reef regions, targeting a total coral deficit area of **96,171 hectares** at a restoration rate of **\$200,000/ha** for a total annual cost of **\$1,923M**.

How it works: Coral nursery infrastructure is deployed at priority reef tracts in each region. Marine labor teams collect coral fragments, maintain nursery structures, and conduct annual out-planting cycles. The Pacific Islands, carrying the largest deficit of **57,371 hectares**, are addressed in parallel at scale.

Who is involved: NOAA Coral Reef Conservation Program, regional governments, marine restoration contractors, local fishing and tourism communities who contribute levy funding under Program 4, and pharmaceutical industry partners with an interest in preserving the biodiversity prospecting pool identified in Section 5.2.

Who is affected and projected future impact: The Gordon-Schaefer model projects 97%+ fishery collapse in all Atlantic regions and 99.8–100% in Pacific regions by 2100 under the status quo. Program 1 is the primary driver of the **\$88.77 billion in NPV savings** by shifting the cover trajectory toward the Restoration scenario. It is also the **only intervention that can arrest pharmaceutical option value losses before they become permanent**: species lost at the near-zero cover troughs of 2028, 2036, 2044, and 2049 are removed from the pharmaceutical pipeline irreversibly, establishing the \$172.6 billion in cumulative pharmaceutical NPV losses as a hard upper bound on additional value at stake. The **500+ million people** relying on reefs for food and livelihoods and the **coastal communities** facing storm damage costs growing from \$380M/year (2025) to \$1,833M/year (2100) are the primary long-term beneficiaries.

Program 2: NOAA Reef Survey Funding — \$5M Annual Total

Model basis: Every model in this paper depends on continuous NOAA CRCP survey data — the SARIMA model cannot be updated without annual benthic observations, the Gordon-Schaefer model cannot recalibrate carrying capacity, Program 4’s parametric triggers cannot be verified, and the species-area curve cannot track pharmaceutical species losses in real time. A gap in NCRMP coverage at a trough year forecloses all option value associated with species lost in that window.

What it is: Sustained annual funding for the NOAA Reef Visual Census (RVC) and National Coral Reef Monitoring Program (NCRMP) survey infrastructure across all seven modeled reef regions [20, 21], now explicitly encompassing biodiversity census components required to maintain pharmaceutical prospecting baselines.

How it works: This program funds the benthic photo-quadrat surveys, fish community censuses, and carbonate chemistry sampling that form the data inputs for all models in this paper.

Who is involved: NOAA CRCP field scientists, regional survey coordinators, and the modeling teams that depend on continuous observational records.

Who is affected and projected future impact: A **single percentage point** of additional coral cover recovered at any region translates to **hundreds of millions** in ecosystem value (e.g., **\$985M present value in Florida per 1% cover**). Making uninformed restoration decisions at \$200k/ha without current survey data risks misallocating the entire \$1,923M annual restoration budget. At \$5M per year, this program functions as the information backbone of the entire six-program package.

Program 3: Fishermen Income Subsidies — \$12M Annual Total

Model basis: The Gordon-Schaefer model projects mean fish biomass reaching critical thresholds before 2060 across all seven regions, with \$4.6 billion in global biomass losses through 2100. These losses fall disproportionately on the approximately 800 licensed commercial fishermen across the seven regions who have no alternative employment base. Program 3 does not prevent collapse — that is the role of Program 1 — but it decouples fishermen’s short-term livelihoods from the collapse trajectory during the 5–10 year restoration horizon.

What it is: A behavior-change program providing **baseline income protection** and **equipment transition support** for commercial fishermen across the seven modeled reef regions.

How it works:

Equipment rebates (50–70%): Subsidize the cost of shifting from reef-dependent bottom fishing to **pelagic species** (mahi-mahi, tuna), which are unaffected by coral degradation. At an average gear transition cost of \$10,000 per fisherman, rebates are estimated at \$4M annually across all regions.

Income disaster payments: Triggered automatically when a fisherman’s annual catch falls **more than 15%** below their **5-year rolling average** — the same threshold used by the USDA for agricultural disaster payments [30]. With ~800 licensed commercial fishermen across the seven regions [7, 21], a 50% trigger rate, and an average payout of \$6,000 per event, income disaster payments total approximately \$2.4M per year. Together with equipment rebates: $800 \times 0.50 \times \$6,000 + \$4M \approx \$12M$ per year.

Who is involved: Regional fishery management agencies, the approximately **800 licensed commercial fishermen** across the seven modeled regions, and the USDA-equivalent disaster payment administration structure.

Who is affected and projected future impact: Over the 10-year program window, the \$120M in total subsidy cost is dwarfed by the \$4.6 billion in biomass losses it partially offsets for the most vulnerable stakeholders, sustaining local communities through the transition period while Program 1 restoration takes hold.

Program 4: National Reef Insurance Reserve — \$8M Annual Total

Model basis: The Gordon-Schaefer stochastic simulations show shock events are the acute mechanism driving cover toward the near-zero troughs of 2028, 2036, 2044, 2049, and beyond. Each event left unremediated for 12–18 months under existing ad hoc funding structures forecloses recovery before the next shock cycle. The permanent pre-discovery extinction component at each trough accumulates irreversibly once species are lost; every month of delayed response translates directly into permanently foreclosed pharmaceutical pipeline value.

What it is: A parametric insurance reserve that makes **automatic, pre-agreed payouts** when coral cover at any monitored reef drops below a defined threshold, without requiring damage assessment or legislative approval. Modeled after the Caribbean Catastrophe Risk Insurance Facility (CCRIF) reef insurance structure [2].

How it works: The reserve is capitalized through levies on commercial fishing licenses (\$15/license annually) and reef-adjacent tourism operations (\$2/visitor annually), funded entirely by the industries that benefit most from healthy reefs. Reserve sizing is derived directly from the Gordon-Schaefer model: with $p_{\text{shock}} \approx 15\%$ annual shock probability and an average coral cover loss of 3–5% per event, a \$50M rapid-response payout per major shock event implies an expected annual payout of \$7.5M, plus \$0.5M administration, for a total of **\$8M** per year. Levy revenue from approximately 267,000 fishing licenses (\$4M) and 2 million annual reef-adjacent visitors (\$4M) funds the full reserve with **no net government outlay**.

Who is involved: National reef management agencies, NOAA CRCP (for parametric trigger monitoring), commercial fishing license holders, reef-adjacent tourism operators, and an actuarial reserve modeling team for annual threshold review.

Who is affected and projected future impact: Program 4 transforms the response timeline from **years to days**, protecting the \$88.77 billion restoration savings scenario from being eroded by event-driven cover collapses before restoration can take hold. The **coastal communities and tourism operators** whose businesses depend on year-over-year reef health are the primary near-term

beneficiaries, alongside patients and pharmaceutical developers whose pipeline value depends on reef species surviving the trough years intact.

Program 5: MPA Monitoring Services — \$6M Annual Total

Model basis: The panel regression in Section 3.3 independently quantifies the mechanism: the marine protection binary coefficient of +1.681% coral cover confirms that enforced MPAs maintain structurally higher cover, directly reducing the harvest term qEX_t in the Gordon-Schaefer model and raising $A(t)$ in the pharmaceutical species-area curve. At \$6M per year, Program 5 is the lowest-cost lever in the package that operates on the same biological mechanisms driving the model's most optimistic recovery trajectories.

What it is: A behavior-change and outcome-modification program that funds **patrol, compliance monitoring, and enforcement** of existing Marine Protected Area no-take zones across all seven modeled reef regions.

How it works: MPA enforcement officers conduct regular patrols of designated no-take zones, issue compliance violations for illegal fishing, and coordinate with regional fishery management agencies to adjust enforcement intensity based on NOAA CRCP monitoring data.

Who is involved: NOAA FKNMS and equivalent sanctuary programs in each region, regional coast guard and enforcement agencies, and environmental NGOs that supplement patrol capacity through collaborative monitoring agreements.

Who is affected and projected future impact: No-take MPAs consistently demonstrate **20–40% higher fish biomass** inside boundaries within **5–10 years** of enforcement, with documented spillover effects benefiting adjacent fished areas. The approximately **800 licensed commercial fishermen** across the seven regions benefit from spillover biomass recovery. By sustaining higher inter-trough cover levels, Program 5 also **reduces the severity of species contractions at bleaching trough years**, partially mitigating the permanent pharmaceutical option value losses quantified in Section 5.2 at an annual cost of just \$6M.

Program 6: Pharmaceutical Biodiversity Prospecting Fund — \$15M Annual Total

Model basis: The pharmaceutical option value model projects \$172 billion in cumulative NPV losses by 2100, with permanent pre-discovery extinction concentrated at the four trough years. Programs 1–5 reduce but cannot eliminate these losses: the Hold Cover and Restoration scenarios reduce trough severity but do not eliminate the permanent component. Program 6 is the **only intervention in the package that addresses this residual, scenario-invariant loss component directly** by banking option value before the trough arrives rather than attempting to prevent the trough itself.

What it is: A **public-private pharmaceutical screening program** that systematically collects, cryopreserves, and screens reef-organism tissue samples from all seven modeled regions before further biodiversity loss forecloses the option permanently. Co-funded on a **50/50 cost-share**, with the government contributing \$7.5M annually and industry matching at \$7.5M annually for a program total of \$15M per year.

How it works:

Cryopreservation banking (\$5M/yr): Tissue and genetic material from priority reef species in each region is collected during NOAA NCRMP survey dives (leveraging the existing Program 2 field infrastructure at no additional mobilization cost) and deposited in a centralized cryogenic biobank. Cryopreservation decouples the discovery probability $p(s)$ from the reef cover trajectory $A(t)$ for banked species, converting at-risk option value into a durable asset that survives bleaching events.

Accelerated high-throughput screening (\$7M/yr): Banked extracts are submitted to existing NIH and pharmaceutical partner screening pipelines for bioactivity profiling against cancer, neurological, and antimicrobial targets — the same therapeutic areas in which reef-derived compounds (AZT, Ziconotide, Trabectedin, Cytarabine) have historically produced approvals. Priority is given to species from the Pacific Islands and Hawaii, which carry the highest species scaling constants c and therefore the largest per-region pharmaceutical option value in the model.

Benefit-sharing agreements (\$3M/yr administration): Legally binding revenue-sharing agreements between the federal government and regional reef communities ensure that commercial royalties from any approved compound are partially returned to the region of origin, creating an enduring financial incentive for local communities to support reef conservation independent of direct government expenditure.

Who is involved: NIH National Center for Advancing Translational Sciences (NCATS), pharmaceutical industry partners (cost-share), NOAA CRCP field teams (sample collection), regional reef management agencies, and local communities as benefit-sharing recipients.

Who is affected and projected future impact: Even if cryopreservation captures only 10% of the species at risk across the four trough events, the expected pharmaceutical option value preserved — on the order of **hundreds of billions of dollars in discovery potential** — substantially exceeds the \$150M ten-year government cost-share of the program. The patients and healthcare systems that depend on future reef-derived treatments are the ultimate beneficiaries, alongside the regional communities that receive benefit-sharing royalties from any resulting commercial approvals.

Total Recommended Annual Funding Package

Program	Govt. (\$M/yr)	Total (\$M/yr)	Scope
1. Coral Reef Restoration	1,923	1,923	Per region (area-scaled)
2. NOAA Reef Survey Funding	5	5	Total
3. Fishermen Income Subsidies	12	12	Total
4. Reef Insurance Reserve	0	8	Reserve (levy-funded)
5. MPA Monitoring	6	6	Total
6. Pharma Biodiversity Fund	7.5	15	50/50 public-private
Grand Total	1,953.5	1,969	

All figures in 2025 USD. Restoration costs assume \$200k/ha (mid-estimate; range \$100k–\$400k/ha). Discount rates: 1.5% (all regions), 80-year planning horizon to 2100. Program 4 carries \$0 net government outlay as levy revenues from fishing licenses and reef-adjacent tourism fully capitalize the reserve. Program 6 government contribution of \$7.5M/yr reflects the 50/50 public-private cost-share; total program spend is \$15M/yr.

When evaluated against the **\$93.07 billion NPV of status quo losses** across fishery, tourism, and wave damage channels, plus the **\$172.6 billion in pharmaceutical option value** at risk under the Section 5.2 model, the \$1,953.5M annual government outlay — if sustained over the 10-year restoration horizon — represents a **positive expected return**: total government cost of approximately \$19.53 billion over 10 years against \$88.77 billion in directly recoverable NPV savings, avoided pharmaceutical pipeline losses, fishery collapse costs, and the banked discovery potential preserved by Program 6 that are not fully captured in the economic loss model's lower-bound estimates. The window for this return remains open — but the near-zero cover troughs projected in 2028 and 2036 establish hard deadlines beyond which pharmaceutical losses become permanent and fishery recovery timelines extend by decades.

Conclusion

Coral reef degradation constitutes a quantifiable, accelerating, and in part irreversible economic risk rather than a distant environmental contingency. Under the status quo trajectory, the integrated modeling framework developed in this paper projects a 77-year net present value loss of \$93.07 billion across wave attenuation and reef-dependent tourism channels, a further \$4.6 billion in fishery biomass losses through 2100, and approximately \$172.6 billion in cumulative pharmaceutical option value losses as episodic bleaching events drive species to local extinction before discovery. The stochastic Gordon-Schaefer simulations estimate collapse probabilities exceeding 97% across all seven modeled regions by 2100, with mean biomass trajectories reaching critical thresholds prior to 2060. These projected losses are not uniformly distributed; they disproportionately affect fishing communities, coastal populations, and Small Island Developing States whose economic stability is structurally linked to reef ecosystems, as well as the patients and healthcare systems whose future treatments depend on compounds that exist only in reef species not yet screened.

The sensitivity analysis demonstrates that the economic consequences of inaction are nonlinear, time-dependent, and channel-specific. For fishery, tourism, and wave damage losses, each year without intervention compresses the potential \$88.77 billion in recoverable savings as coral cover approaches its low-cover asymptote — but these losses remain theoretically recoverable under active restoration. For pharmaceutical option value, the dynamic is categorically different: losses at the episodic bleaching trough years are permanent, accumulating more than \$50 billion in irreversible pre-discovery extinction value that no subsequent restoration can reclaim. This distinction is the central actuarial finding of this paper: the risk is not merely large but partially uninsurable after specific threshold events, making the timing of intervention as consequential as its scale.

The modeled mitigation package totaling \$1,953.5 million in annual government outlay therefore represents a risk management strategy grounded in quantified expected value reduction across all four loss channels. Programs 1 through 5 address the recoverable fraction of losses through restoration, monitoring, behavioral change, and parametric insurance. Program 6 addresses the irreversible fraction directly by banking pharmaceutical option value before trough years foreclose it permanently. When evaluated against the compounding trajectory of projected losses across all channels, the six-program package yields a positive expected return and preserves the structural ecosystem services upon which coastal economies and future medical discovery depend.

The results of this analysis indicate that the window for economically efficient intervention remains open but is narrowing on two distinct timelines. The recoverable losses from fisheries, tourism, and wave damage demand restoration investment before the mid-2050s critical threshold. The irreversible pharmaceutical losses demand cryopreservation action before 2028. Policymakers, insurers, and regional governments must treat these as separate but equally urgent imperatives: proactive investment in reef resilience is not solely an environmental objective but a necessary component of long-term economic risk mitigation, coastal financial protection, and the preservation of a pharmaceutical discovery frontier that, once lost, cannot be recovered at any price.

Appendix

Appendix 1: ARIMA and SARIMA Candidate Model Selection

Model	AIC	BIC
AR(1)	48.358	49.551
MA(1)	45.971	47.165
ARMA(1,1)	50.165	51.757
AR(2)	50.357	51.949
MA(2)	46.580	48.172
ARMA(2,1)	48.090	50.079
ARMA(1,2)	48.401	50.390
ARMA(2,2)	44.770	47.158
White Noise	46.822	47.617
SAR(1)×(1,4)	42.645	44.237
SAR(1)×(1,5)	49.520	51.112
SARMA(1,1)×(1,1,4)	39.409	41.796

Appendix 2: SARMA and SARIMAX Hard Coral Cover Forecasts (Selected Years, 2024–2100)

Year	SARMA Forecast (%)	Lower 95% (%)	Upper 95% (%)	SARIMAX Forecast (%)
2024	6.09	3.72	6.95	9.24
2043	4.97	1.53	6.24	6.01
2062	3.86	1.26	5.23	3.52
2081	4.43	1.51	6.05	3.26
2100	3.85	1.28	4.53	1.73

Appendix 3: Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted in MATLAB to visualize trends, identify variability, and detect anomalies prior to modeling. Figures 13 and 14 illustrate pre-ARIMA patterns. EDA confirmed an inverse relationship between hard coral and algae cover, bleaching-related disruptions in the benthic time series, and a long-term upward trend in sea surface temperatures, supporting the inclusion of seasonal components in the ARIMA model and validating the thermal stress assumptions embedded in the bleaching framework.

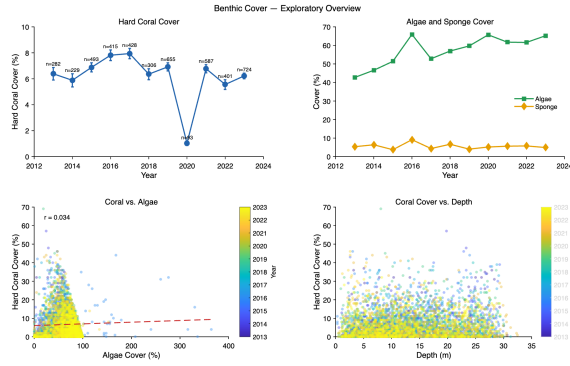


Figure 13: Pre-ARIMA benthic cover visualization. Algae rises as coral cover declines, with hard coral near 6%.

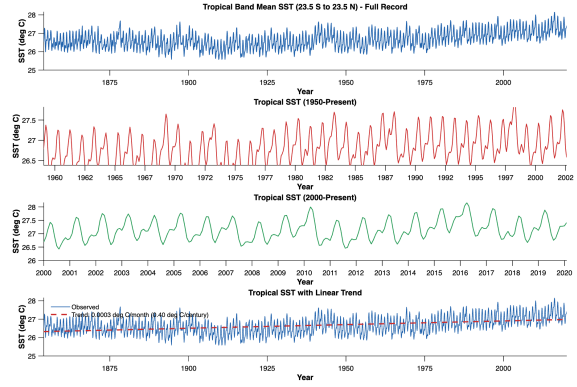


Figure 14: Pre-ARIMA water temperature visualization showing a steady SST increase in tropical regions.

Appendix 4: Components and Variables of the Gordon-Schaefer Model

#	Component	Variable	Significance
1	Fish	K	Fish population carrying capacity
2	Fish	x_0	Most recently measured biomass
3	Fish	r	Fish population growth rate
4	Fish	q	Catchability coefficient - measures a fishery's efficiency in catching fish
5	Fish	σ_x	Coefficient of variation of fish population - used as the diffusion coefficient in stochastic component
6	Fish	E	Effort units - measures the tendency of a fishery to put resources into fishing & difficulty to obtain harvestable yield
7	Reef	A	Reef area
8	Reef	$R_{\max}$	Maximum observed reef hard coral cover
9	Reef	R_0	Most recently observed reef hard coral cover
10	Reef	g	Reef recovery rate
11	Reef	δ	Reef degradation rate
12	Reef	σ_R	Coefficient of variation of reef degradation - used as the diffusion coefficient in stochastic component
13	Reef	α	Ratio of biomass to % coral cover
14	Economic	p_{fish}	Average price per kg of fish
15	Economic	d	Discount rate - accounts for inflation as losses in the future are less impactful than present ones
16	Shock	p_{shock}	Probability of a shock occurring
17	Shock	μ_{reef}	Average coral cover loss due to a shock
18	Shock	σ_{reef}	Standard deviation coral cover loss due to a shock
19	Shock	μ_{fish}	Average fish population loss due to a shock
20	Shock	σ_{fish}	Standard deviation fish population loss due to a shock

Appendix 5: Components and Parameters of the Economic Loss Model

#	Component	Variable	Significance
1	Tourism	$T(A)$	Annual reef-dependent tourist arrivals at coral cover A
2	Tourism	T_0	Baseline tourist arrivals at reference cover; $T_0 = 2.10\text{M}/\text{yr}$ [14]
3	Tourism	A_0	Reference hard coral cover level; $A_0 = 6.39\%$ [16, 18]
4	Tourism	M_0	Average economic output per tourist; $M_0 = \$1,800/\text{person}$ [6, 15]
5	Tourism	M_{base}	Baseline annual tourism revenue; $M_{\text{base}} = T_0 \cdot M_0 = \$3,780\text{M}/\text{yr}$
6	Tourism	$\Delta M(t)$	Annual tourism loss relative to baseline
7	Wave	E_0	Offshore incident wave energy [$\text{kJ m}^{-1}\text{yr}^{-1}$]
8	Wave	k	Bulk wave attenuation coefficient [per % cover per km^2] [5]
9	Wave	$C(t)$	Hard coral cover at time t [%]
10	Wave	A_{reef}	Reef plan-form area [km^2]; $A_{\text{reef}} = 9.0 \text{ km}^2$ [19]
11	Wave	D_{max}	Maximum annual coastal damage cost; $D_{\text{max}} = \$850\text{M}$ [24, 29]
12	Wave	λ	Damage saturation rate; $\lambda = 2.5$ (calibrated)
13	Wave	ΔE_{frac}	Fractional excess shore energy above healthy-reef baseline
14	NPV	r	Social discount rate; $r = 1.5\%$ per year [32]

GitHub Repository

The full data processing scripts, model code, and generated figures are archived in the project GitHub repository. The top-level structure follows a standard workflow layout: **data/** contains raw and cleaned datasets by region, **notebooks/** contains the code used to clean the data, **outputs/** contains exported plots used in this report, and **MATLAB/** contains the code used to simulate the models. The repository can be found at <https://github.com/coreylu2027/CAAAARal-Reefs/tree/main>.

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